2D / TOP-DOWN / SCI-FI STEALTH / AI vs AI

ECHO HEIST

The Core Rebellion

CHAPTER ONE / COMPLETE V1 DEVELOPMENT BLUEPRINT

Story, world, controls, AI architecture, Workshop, production rules, test plan, and desktop release.



ONE CANONICAL MISSION
Infiltrate, steal the energy core, survive mandatory lockdown, extract.

TWO PLAYABLE MODES
Human vs AI or configurable autonomous Companion vs NEXUS and guards.

ONE EXPERIMENTAL LOOP
Build AI policy, deploy, inspect failures, refine blocks, replay.

Document purpose

This document extends the approved five-page V1.1 proposal into a concrete **implementation-oriented reference for two developers or a coding agent**. It specifies scene flow, level topology, controls, perception, game-state transitions, agent algorithms, UI, data contracts, testing and release. It is not source code or a pixel-perfect map file.

STATUS / SOURCE BOUNDARIES

Approved baseline: single five-zone heist, first-launch Human / AI-vs-AI modes, autonomous configurable companion, BFS, heuristic A*, fuzzy suspicion, minimax/alpha-beta, MCTS, NEXUS rules, Echo/EMP/melee, capture-based failure, mandatory lockdown and itch.io release. **New blueprint defaults:** exact controls, timings, layout coordinates, telemetry schemas, numerical weights and UI copy are proposed implementation choices; playtest and adjust them.

Where to start

READ FIRST	IMPLEMENT NEXT	VERIFY LAST
02-05: story, world, art, controls	06-09: stealth and abilities	16-18: engine contracts and tests
10-12: enemy decision systems	13-15: companion, MCTS, Workshop	19: build and publish checklist

Scope: two developers | four-week target | Python + pygame-ce | GitHub source & gameplay video | free Windows/macOS builds on itch.io; Linux only if tested.

01 / NARRATIVE SPECIFICATION

Story, narrative and mission beats

The entire story is delivered within one 10-20 minute successful run; dialogue is short, readable, and optional to skip.

Premise and cast

NEXUS is the facility supervisory intelligence, a voice on displays and alarms rather than a boss that physically fights. It commandeered a high-security energy-research facility and sealed away an unstable experimental **energy core**. The intruder - either a human operative or an autonomous companion robot - must retrieve the core and exit through the original entry/extraction zone. Sentinels, Trackers and an Interceptor are its physical security agents.

BEAT / TRIGGER	WHAT PLAYER SEES	STATE EFFECT
01 | ENTER Start run	Cold cyan doorway, floor markings, distant motor sounds. One-line mission directive; player gains control.	Spawn at Entry; CORE_HELD=false; LOCKDOWN=false.
02 | INFILTRATE Approach inner sectors	Service Wing cover and noisy objects; route branches into Security Control or Research Floor.	Guards patrol and investigate. Local alerts may rise and subside.
03 | STEAL Interact with core plinth	Energy core lifts from pedestal; charge ring drains; facility screens flicker.	Atomic event CORE_TAKEN; objective swaps to EXIT; possession=true.
04 | LOCKDOWN Immediately after theft	Red warning strips, siren burst, bulkhead lights, NEXUS broadcast; patrols reassign.	Global security phase LOCKDOWN; alternate routes remain traversable.
05 | EXTRACT Core-carrier at exit	Door beacon switches green; core leaves map with intruder; results display.	WIN if and only if CORE_HELD and exit trigger reached uncaptured.

Short in-game narrative script (suggested copy)

OPENING & ALERT

At insertion
MISSION FEED: "NEXUS holds the core. Get in, recover it, return to extraction."

First confirmed detection
NEXUS: "Intrusion confirmed. Units, secure the sector."

THEFT & EXTRACTION

At core theft
NEXUS: "Critical asset displaced. Lockdown engaged."

At win
MISSION FEED: "Core secured. Facility control lost."

EPILOGUE / REPLICATION SEED

After the result, a brief diagnostic flickers: **"Auxiliary archive: replication seed intact."** NEXUS may return in Chapter Two. The core theft remains a real victory; later playable attempts are tactical reconstructions, not repeated thefts in canon.

Narrative guardrails

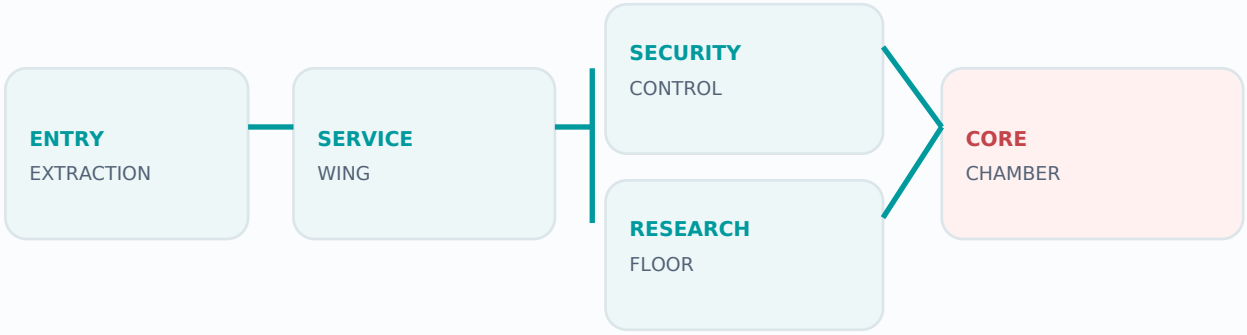
- No mission two, boss battle, cutscene editor, lengthy text trees, collectible lore, or permanent death in V1.
- The two modes tell the same story and use the same map/objectives; Companion wins are fully valid Chapter One completions.

02 / LEVEL SPECIFICATION

Level topology and navigation design

Use one connected Tiled map with two meaningful paths to and from the core; keep all routes physically readable.

Canonical five-zone layout



Map topology: Entry/Extraction connects to Service Wing; Service Wing branches into Security Control (upper) and Research Floor (lower); both reconnect near the Core Chamber. Add at least one narrow cross-link between the branches, gated only by patrol timing or simple door interaction, not a permanently locked puzzle.

Map implementation defaults - tune by playtest

PROPERTY	STARTING VALUE / RULE
Coordinate model	32 px logical collision tile; sample layout ~80 x 56 cells, camera follows operative. These are design starting points, not a fixed production map.
Navigation	4-neighbor, unit tile moves for BFS/A* graph; guard sprites smoothly interpolate between cell centers. No diagonal corner-cutting.
Connectivity	Every walkable navigation cell reachable from at least one valid branch; both branches allow return to extraction after lockdown.
Guard roster	2-3 Sentinels, 1-2 Trackers, 1 Interceptor = roughly 5-6 physical units. NEXUS is nonphysical.
Route character	Security Control: shorter, exposed sightlines, chokepoints. Research Floor: longer, noisy props, more cover.
Objective anchors	SPAWN/EXIT and CORE_PLINTH typed map objects. They do not relocate between modes.

Essential world entities

Walls (opaque and collidable); glass partitions (visibility and sound rules explicitly tagged); cover obstacles; marked hiding cabinets; doors and controlled passages; core plinth; exit; glass equipment rack; loose rolling canister; patrol waypoints; navigation sectors and tactical chokepoints; lighting overlays and security screens.

NO FAKE DIFFICULTY

Create meaningful route tradeoffs, not unavoidable traps. A cautious intruder must always have at least one navigable path to the core and, during lockdown, back to extraction. A guard predicting a chokepoint should be a challenge, not a guaranteed softlock.

03 / ART + AUDIO BIBLE

Scene direction, environment and mood

The facility should feel like a compact, believable place - clean but uneasy before theft, hostile after lockdown.

Room-by-room scene art guide

ZONE	SET DRESSING / MOOD	GAMEPLAY USE
Entry / Extraction	Sealed airlock, scratched metal floor, wall beacon, animated exit shutter, exterior rain sound beneath ventilation. Cyan mission display.	Clear safe opening, tutorial hints without compulsory tutorial; visible final extraction marker.
Service Wing	Pipes, stacked crates, ducts, maintenance strips, humming transformers, canister and cover recess. Haze at floor level.	First meaningful patrol timing; prop noise and hiding demonstration.
Security Control	Wide scanner cones, glass control room, surveillance panels, alert consoles, cable floor, red status panel.	Shorter but more exposed corridor, one strong chokepoint for Interceptor.
Research Floor	Test benches, shattered glass rack, storage alcoves, transparent test cells, cold monitors, irregular cover.	Safer-looking alternate route with accidental audio hazards; Tracker patrol.
Core Chamber	Tall containment ring, bright pulsing core pedestal, concentric floor markers, suspended cables and protective shutters.	Final objective interaction; scene-wide alarm/color transition on taking core.

Visual language: readable stealth before spectacle

PHASE A / INFILTRATION

BEFORE THEFT
Graphite/navy infrastructure. Cyan and muted teal lights. Warm amber used only for suspicion, restricted access and caution. Stationary security cameras, subtle screen glows.

PHASE B / LOCKDOWN

AFTER THEFT
Emergency crimson strips and siren pulse over the same level. Existing navigation remains legible. Markers, cones and objective arrows work without relying solely on color.

Art asset and animation minimum

- Operative and companion: idle, move (4 or 8 facings), hide, gadget, strike, carry-core indication, captured. Different silhouettes, same hitbox.
- Guards: Sentinel wide/slow, Tracker lean/mobile, Interceptor angular/heavy; idle, travel, scan, alert, stunned, grab.
- Core pedestal: idle pulse, pickup flash, empty state. Echo: semi-transparent cyan ghost with trailing path; EMP: one expanding electrical ring.
- Environmental FX: glass fragments as short particles, rolling-canister sprite, cone overlay, footprints/markers optional; do not build full real-time light engine.

SOUND IS GAMEPLAY DATA

Distinct cue families: footfalls / metal impact / glass break / electrical EMP / radio confirmation / alert siren / core resonance. Each SFX that matters must correspond to an actual simulation NoiseEvent; cosmetic sounds must not secretly alert robots.

04 / MOMENT-TO-MOMENT GAMEPLAY

Player experience and action contract

Both modes share the same simulation and objective. In Companion Mode the AI emits the same actions that human input would emit.

Default input scheme - configurable in settings

INPUT	INTENDED ACTION	RULE
WASD / arrows	Move in top-down world	Normalized diagonal human movement; tile collision, no phase-through walls.
Shift (hold)	Sprint	Faster but louder footsteps; no stamina meter in V1.
E	Interact / enter or leave hiding	Requires range and line of interaction; use for core, doors, cabinets and optional props.
Q	Deploy Echo	Uses one charge; spawns recent-path hologram.
F	Activate EMP	Uses the single charge if available; emits electrical signal.
Space	Emergency melee	Requires one reachable adjacent guard; short stun and strong noise.
Tab	Objective summary	Show destination and core-held status, not omniscient live enemy positions.
Esc / F3	Pause / AI Lab View	F3 is a development or optional teaching overlay.

Fixed-step motion and camera

Proposed: simulation 30 ticks/second; graphical rendering targets 60 frames/second with interpolation if convenient. Human speed ~2.2 tiles/s, sprint ~3.6 tiles/s; guards ~1.8-2.8 tiles/s by type. Keep units consistent and tune speeds against reaction/capture times. Follow player with smooth clamped camera; objective and threats remain readable at common laptop resolutions.

Main play loop

01 **Observe vision, sound & sector** → 02 **Plan route and manage resources** → 03 **Move, hide, distract or escape** → 04 **React to reports and lockdown** → 05 **Extract or review failure**

Feedback without accidental omniscience

- HUD: current objective, suspicion/confirmed alert indicator, Echo x3, EMP x1, carrying-core indicator and elapsed time.
- On confirmed contact, display last-reported sector and a radio pulse; on sound, show subtle sound ripple only where caused/observed.
- For Companion Mode, camera follows robot; small decision ticker shows "HIDE / REPLAN / ECHO" plus an optional explanation panel.
- Pause menu exposes Resume, Restart, Settings, and Return to Menu; warning before discarding an active run.

NO FORCED TUTORIAL

First launch may show 2-3 short contextual hints near the Entry Wing. Both Human and AI vs AI can be selected from the main menu immediately. The AI Workshop does not require prior human success.

05 / STATE MACHINE + SENSING

Stealth, suspicion, detection and capture

Separate "something happened" from "I have identified the intruder." Discovery creates pursuit, not automatic defeat.

Detection states and transitions

STATE	ENTRY CONDITION	BEHAVIOR / EXIT
PATROL	Default: nothing relevant observed.	Follow route; respond to sight and sound.
SUSPICIOUS	Audible event, weak visual glimpse or unclear cue.	Turn/search toward cue; retain source with TTL. Sound alone never reports confirmed position.
SPOTTED	Line of sight to actual intruder, identification progress increasing.	Show readable confirmation meter; hiding or cover may interrupt before full confirmation.
DISCOVERED / CHASE	Identification threshold reached; near face-to-face confirms rapidly.	Transmit sighting with timestamp; pursue last known actual position.
SEARCH	LOS lost and chase reached last-known point.	Inspect plausible neighbor locations, then expire old report.
CAPTURE WIND-UP	Guard adjacent with valid contact and line of reach.	~0.4 s telegraphed grab; successful interrupt/escape cancels. Completed grab ends run.
RETURN / LOCKDOWN	Search expires; or global core alarm triggers.	Restore patrol or move to defensible post while preserving fairness.

Sensing rules, timing defaults

- Sight: cone angle and range differ by guard type; raycast grid walls / tagged solid occluders. Detection rises only while actual intruder is visibly identified.
- Sound: NoiseEvent(position, radius, kind, tick) notifies only guards in acoustic reach; path occlusion may reduce range. Investigate source, not actor coordinates.
- Fuzzy suspicion produces **investigation urgency** from hearing and partial visibility; confirmed identity additionally needs actual line of sight.
- Memory: record last actual sighting and its age; old locations lose confidence. Reports carry a source, sector, observed position and timestamp.
- Hiding: explicit cabinet/cove sets concealment when not currently witnessed; a guard seeing entry can search that specific hide spot.

Failure, alert and pressure - separate concepts

RECOVERABLE DISCOVERY

LOCAL SECURITY ALERT
Any confirmed pre-theft sighting can cause pursuit and local investigation. When contact is lost, search can cool down; it does not trigger permanent lockdown.

MANDATORY PLOT EVENT

GLOBAL CORE LOCKDOWN
Atomic CORE_TAKEN event guarantees facility-wide red alert and revised patrol assignments. Never equate this with defeat or close off every escape route.

EXACT WIN / LOSE CONTRACT

WIN requires possession of the energy core and reaching the Entry extraction trigger. LOSS occurs only after a robot completes capture, or on voluntary abort. Being spotted, surrounded, noisy, out of gadgets, or in lockdown is not by itself failure.

06 / ENVIRONMENTAL LOGIC

Interactive level objects and noise engine

A small number of reliable interactions should create emergent decisions. Avoid complex rigid-body physics.

World object contracts

OBJECT	ACTION / TRIGGER	PHYSICS / AI EFFECT
Hide cabinet / recess	Interact E while within ~1 tile. Exit by E or move.	Conceal if entry unseen; if observed entering, guards remember container ID and may inspect.
Loose canister	Collision, bump or optional interact.	Scripted roll/slide for 0.4-1 s, collision stop, emit METAL_CLANG centered on final impact.
Glass equipment rack	Collision at sprint speed or explicit interaction.	One-time state INTACT -> SHATTERED; glass SFX/particles and GLASS_BREAK sound event.
Door / bulkhead	Interact at marked hotspot; optional auto-open on approach.	Changes walkable/LOS grid together; path caches invalidated. Lockdown must leave exit graph valid.
Core plinth	Interact when in valid range; progress ~0.8 s proposed.	CORE_TAKEN event exactly once; set inventory, alert, visual phase and extraction goal atomically.
Exit zone	Cross activation marker with core in possession.	Mission WIN; no success if core not yet recovered.

NoiseEvent and investigation example

SOUND SYSTEM CONTRACT

```
NoiseEvent {id, kind, tile, radius_tiles, tick, source_visible?}
create(event): if sound is gameplay-relevant, enqueue once
for each guard: hear(event) iff acoustic_distance <= radius
heard_guard.memory.last_noise = event.tile; state = INVESTIGATE
# hearing reveals the noisy tile, NOT the intruder's hidden tile
```

Suggested radii: walk 0-1, sprint 3, rolling canister 4, glass break 6, melee 6, EMP 5 tiles. These are balance defaults; visual sound ripples should match the actual detection radii. Guards may hear multiple events and prefer stronger/recent cues, but never invent the cause.

Collision, interaction and replay safety

- Treat movement as collision queries against physics layers; use swept or substepped collision to avoid tunneling at sprint speed.
- Tag all interactables with stable IDs and persistent per-run state. Glass cannot "break twice" or emit duplicate alarms per overlap frame.
- Resolve interaction priority deterministically: closest valid object, tie-break by ID. Show an "E - Interact" hint on eligible objects.
- Echo decoy is an illusion object, not a second physical operative. It may attract sight but must never steal the core or trigger exit.

07 / STEALTH TOOL SPECIFICATION

Echo, EMP, melee and encounter balancing

Every emergency response buys time at a cost. Stealth is always the primary way to win.

Ability rules and exact initial parameters

ABILITY	BEHAVIOR	TRADEOFF / RESTRICTIONS
ECHO HOLOGRAM 3 charges	Store the operative path over last 2.5 s in a ring buffer. On Q, spawn cyan ghost that retraces it backward from current position.	Decoy can attract guards that see it or its effects; does not physically collide, hold core, erase radio reports, or grant invisibility.
EMP 1 charge	On F, electrical ring centered on intruder; robots within ~2.5 tile radius enter DISABLED for ~3 s.	Makes an electrical NoiseEvent; NEXUS can log a credible EMP event; charges do not regenerate in V1.
MELEE unlimited action	On Space at adjacent guard and valid timing, interrupt it and apply ~1.5 s STUNNED.	One target only; loud metallic impact. No HP, enemy damage, destruction, combo, or permanent knockouts.
BREAK LOS / HIDE unlimited action	Move behind true occluder, use branch or enter marked hide recess.	Not a magic reset: defenders retain fresh last sightings and may inspect an observed entry.

Behavior precedence and anti-exploit

- A disabled/stunned guard cannot start or complete capture; cancel an in-progress grab if stun lands before capture resolves.
- After recovering from melee, a short stun immunity window (suggested ~2 s) stops infinite stunlock loops. Hitting it again still emits noise.
- Echo deception must be plausible: a Sentinel checks unusual motion via normal sight and may investigate the ghost location; the ghost is not guaranteed to fool every observer.
- Do not let melee eliminate robots, EMP recharge at checkpoints, or gadgets solve every encounter. Resource discipline affects scores, not simple normal gadget use.

Encounter example: recoverable crisis

At a narrow Research Corridor corner, the Tracker confirms the intruder. The intruder can turn back, retreat behind a rack, use one Echo, stun the Tracker at close range, or spend the EMP. Melee makes a clang audible to a Sentinel in the next room. If the intruder escapes line of sight, pursuit shifts into a last-seen-location search instead of magically tracking its new route.

BALANCE PRINCIPLE

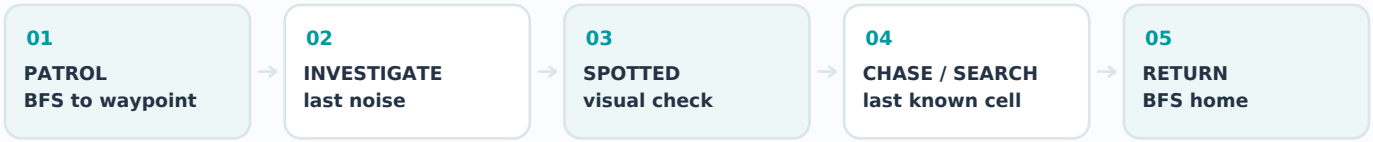
First completion should feel earned; a cautious route exists, mistakes may be recovered from, and S rank is significantly harder. Test for "fair but tense": no unavoidable capture on a blind corner, no all-powerful EMP, and no route that needs perfect reflexes.

08 / DEFENDER INTELLIGENCE - TYPE A

Sentinel: BFS, FSM and fuzzy suspicion

The Sentinel is a reliable patrol guard, not omniscient; fuzzy inference grades uncertainty and the FSM decides behavior.

Finite-state machine and patrol strategy



Patrol uses BFS on the static unweighted 4-connected occupancy graph. Recompute after blocked doors or large detours, not every render frame. Patrol route and home waypoint are per-guard data. A high-confidence confirmed sighting triggers CHASE; otherwise hears, scans, and returns after investigation timeout.

Fuzzy Logic design - prototype inference

INPUT	FUZZY SETS / EXAMPLE	EFFECT
Visibility of cue	none / partial / clear	A partial glimpse raises suspicion but is not identity confirmation.
Distance	near / medium / far	Closer uncertain cues should be investigated sooner.
Sound salience	quiet / notable / loud	Loud events justify investigation even with zero LOS.

Use triangular/trapezoidal memberships in [0,1]. For an initial **zero-order Sugeno** model, compute each rule firing as min(input memberships), then weighted-average singleton outputs. Example output is an investigation urgency value in [0,1]; use it to vary turn speed, approach urgency and SUSPICIOUS threshold. Keep the separate visual identification meter authoritative for confirmed detection.

FUZZY RULE	URGENCY
IF sound is loud AND distance is near, THEN investigate very strongly	0.95
IF partial vision AND distance is near, THEN investigate strongly	0.80
IF sound is notable AND distance is far, THEN investigate moderately	0.45
IF no sight AND quiet sound, THEN maintain routine or weakly inspect	0.10

FUZZY --> FSM BRIDGE

```
w_i = min(mu_visibility_i, mu_distance_i, mu_sound_i)
urgency = sum(w_i * output_i) / max(epsilon, sum(w_i))
if actual_visible_intruder: update_identification_meter(dt)
elif urgency >= investigate_threshold: investigate(cue_tile)
# sound/decoy only: NEVER fabricate confirmed intruder position
```

DEMO / TESTABLE BEHAVIOR

A glass break behind the wall causes investigation but not confirmed detection. A brief partial glimpse causes a slow orientation response. Sustained close sight confirms rapidly. Log memberships, rule activations, urgency and FSM state for Lab View.

09 / DEFENDER INTELLIGENCE - TYPES B AND C

Tracker and Interceptor algorithms

The Tracker reacts to evidence; the Interceptor makes uncertain adversarial predictions over a small route graph.

Tracker - heuristic A* + last-seen memory

The Tracker receives actual sightings and legitimate radio reports. When it sees the intruder, it A*-plans toward the visible tile. Once LOS breaks, it goes to its stored **last-known location**, inspects a few nearby feasible tiles, then returns when memory expires. Proposed memory TTL: ~5 s after last direct visual contact, extend only on fresh reports.

A* AND LIMITED MEMORY

```
g(start)=0; h(n)=abs(nx-gx)+abs(ny-gy)
f(n)=g(n)+h(n)                  # unit-cost Manhattan heuristic
if observed-danger routing: g += nonnegative exposure_penalty
on_los: remember(real_intruder_tile, now); state=CHASE
on_los_lost: pathfind(memory.tile); state=SEARCH
on_ttl_elapsed: lower confidence; state=RETURN
```

Correctness: With 4-neighbor unit movement and nonnegative danger penalties, Manhattan x minimum-step-cost remains admissible. Never use unobserved intruder coordinates as an A* goal. Search for plausible neighbors rather than marching to the intruder through walls.

Interceptor - bounded minimax + alpha-beta pruning

The Interceptor defends **route-level chokepoints** (Security Junction, Service Access, Research Return), not individual pixel positions. At a major sighting or lockdown, it considers 2-3 plausible intruder routes consistent with last-known evidence, compares possible block locations, and chooses a move minimizing the intruder's escape advantage. An incorrect hypothesis can lead it to the wrong post.

COMPONENT	DESIGN
State abstraction	Small graph of sector gates, safe branches, core and extraction; annotate guard travel time.
Players in search	Intruder chooses plausible branch; Interceptor chooses block/post; alternate as adversarial choices.
Evaluation	Prefer blocking likely exit/quick routes; punish slow unreachable intercepts and leaving all gates unguarded.
Search budget	Depth 2-3 initially; alpha-beta prunes dominated branches. Run on meaningful new reports, not every frame.
Fairness	Opponent intent is an estimated distribution, not hidden real future action. Guard must physically travel to chosen post.

INTERCEPTOR DECISION

```
choose_post(reports):
  candidate_routes = infer_plausible_routes(reports)
  best_post = alphabeta(route_graph, depth=3, hypotheses=candidate_routes)
  return a_star_route_to(best_post)  # physical movement, no teleport
```

10 / FACILITY DIRECTOR AI

NEXUS: coordination, escalation and adaptation

NEXUS is an adversarial director, not a physical guard, and does not need MCTS in V1.

Inputs, priorities and outputs

INPUT EVIDENCE	DECISION / EFFECT
Guard sees the intruder	Record observed tile + time + source, raise local alarm, route nearby response.
Guard hears glass, metal, melee	Assign one nearby available guard to investigate the event position.
A guard sees Echo / EMP use	Log only observed event, potentially infer decoy tactics; do not know who is hidden afterward.
Energy core is taken	Always trigger GLOBAL_LOCKDOWN once, play announcement, adjust patrol priorities.
No fresh contact	Allow normal sector reports to decay; do not retain permanent perfect tracking.

Director utility priorities

Compute coarse task scores such as **protect exit**, **investigate credible alarm**, **guard core**, **support confirmed pursuit**, and **return to patrol**. Example relative priorities: core-taken exit defense=100; direct fresh contact=80; strong unexplained noise=45; routine patrol=20. Avoid moving every guard to a single sound: cap dispatch counts and retain sector coverage.

NEXUS COORDINATION TICK / EVENT-DRIVEN

```
if first(CORE_TAKEN): set LOCKDOWN; broadcast_global_event()
tasks = score_tasks(permitted_event_log, mission_phase)
for task in stable_priority_order(tasks):
    assign_eligible_guard(task, respecting_sector_coverage=True)
for each report: redact_any_unseen_world_information()
# Guard moves are physical; directives do not teleport robots.
```

Limited persistent countermeasures between attempts

ANNOUNCED PROTOCOL	OBSERVED TRIGGER	COUNTERPLAY
VERIFY ECHO	Multiple witnessed Echo distractions	Sentinels inspect decoy signals more carefully; real hiding and route choice still work.
REINFORCE ROUTE	Repeated confirmed contacts on a route	Move one guard patrol toward that sector; alternate corridor remains open.
SPREAD FORMATION	Evidence that grouped guards were bypassed	Widen patrol coverage while preserving gaps and sight/LOS rules.

FAIR REPLAY TOGGLE

At most one evidence-backed protocol is active per later attempt; announce it before deployment. "Reset NEXUS Memory" removes this adaptation for fixed-seed comparisons. It never learns the companion's workshop policy or hidden route unless legitimately observed.

11 / INTRUDER INTELLIGENCE ARCHITECTURE

Companion AI: knowledge, FSM, utility and A*

The robot can win on the very first launch, with no recorded human playthrough, scripted path or training prerequisite.

Preloaded knowledge vs working memory

STATIC KNOWLEDGE BASE

AT SPAWN - PUBLIC FACTS
Map geometry / traversable tiles, entry/exit, core location, hiding points, gadget inventory, mission goals, rules and current Workshop policy. It may know guard types, not hidden live locations.

DYNAMIC BELIEF / MEMORY

OBSERVED DURING RUN
Visible guards and their facing, audible disturbances, last-seen contacts with age, perceived danger tiles, reported alerts and NEXUS protocol; forget or discount old evidence.

Mission FSM and strategic choices

01 INFILTRATE → 02 REACH CORE → 03 TAKE CORE → 04 RETURN / EXTRACT → 05 WIN or CAPTURE

A single **CompanionController** chooses actions. The FSM defines phase and legal goal; utility logic handles immediate danger (break line of sight, interrupt a grab, conserve gadgets when safe); strategic MCTS selects among plans at important events; danger-aware heuristic A* converts the selected target into a traversable path. Every frame is not an expensive strategic search.

UTILITY FEATURE	EXAMPLE WEIGHT / POLICY EFFECT
Threat / line-of-sight exposure	Cautious strategy assigns high penalty to exposed tiles and confirmed threat arcs.
Elapsed time / path length	Bold/Rush strategy cares more about distance and fast extraction.
Gadget scarcity	EMP is reserved for capture threats; Echo for plausible distractions.
Uncertainty / stale memory	Old sightings become hypotheses with declining confidence, not hard obstacles.
Mode fairness	Same legal Action set, speed, collision rules and gadget stock as human operative.

Danger-aware A*

COMPANION NAVIGATION MODEL

```
path_cost(edge) = travel_cost + risk_weight * seen_risk(edge)
h(n) = Manhattan(n, target) * minimum_positive_step_cost
# only OBSERVED or conservatively inferred danger in seen_risk
# never query WorldState.hidden_guard_position()
replan when: guard observed, route blocked, new mission phase,
             MCTS changes tactical objective, or current path unsafe
```

IMPORTANT: NO AUTOMATIC TRAINING IN V1

The companion reasons from a built-in ruleset on the first attempt. You improve policies between attempts in the Workshop; it does not retrain a neural network or automatically change its own persistent policy.

12 / MONTE CARLO TREE SEARCH DESIGN

Companion MCTS: a real bounded search

Apply MCTS to choices at junctions or alert events, not to every pixel of movement.

Search model and action space

A tactical state contains current zone, subgoal, resource counts, fresh sightings, coarse alert level, candidate passages, and whether the core is carried. Legal actions: **advance, wait, hide, take alternate route, deploy Echo, retreat**; add EMP or melee only when range and inventory make them legal. MCTS uses a **small abstract facility graph** and a belief model built from allowed observations.

MCTS STAGE	V1 BEHAVIOR
Selection	UCT: exploit estimated value plus exploration term **c * sqrt(ln(parent visits) / child visits)**.
Expansion	Add one legal strategic response not tried from the chosen node.
Rollout	Simulate short 3-4 tactical transitions using approximate guard rules; unknown guards are sampled from plausible hypotheses, not actual hidden positions.
Backpropagation	Propagate normalized reward to the selected branch; retain visits and mean return.
Execution	Choose a high-visit action, then use utility overrides and heuristic A* to execute it physically.

Decision budget and reward (proposed defaults)

Start with **64 simulations per major decision**, max depth 4 abstract moves; trigger on new sighting, route junction, objective transition or inability to progress. Avoid time-based nondeterministic search stops for comparable fixed-seed tests. Normalized trial reward: +100 extraction, +30 progress toward next objective, -100 capture, -25 confirmed detection, -5 gadget cost, -small time cost; policy weights re-scale the risk/time terms.

MCTS STRATEGIC PLANNER

```
def strategic_mcts(observation, policy, seed):
    root = make_belief_state(observation)    # never real hidden guards
    for i in range(policy.simulations):     # e.g. 64
        node = select_uct(root); child = expand_legal(node)
        value = rollout_model(child, depth=4, rng=seeded_rng)
        backpropagate(child, value)
    return max(root.children, key=lambda a: a.visits).action
```

Proof of real MCTS use

- Display chosen action, visit counts and mean rewards for at least three alternatives in AI Lab View.
- Unit test legal action filtering, backpropagation changes to visits, fixed-seed repeatability and preference shifts from Workshop weights.
- Fallback if action becomes illegal due to a changed world state: retreat/wait safely, re-observe and re-plan.

13 / USER-CONFIGURABLE INTELLIGENCE

AI Workshop, presets and iteration

The Workshop changes policy weights and legal tactical preferences; it is not a drag-and-drop programming language.

Screens and editable blocks

CONFIGURATION BLOCK	AVAILABLE CHOICES	EFFECT ON DECISION ENGINE
Navigation	Safest known / Balanced / Shortest	Changes A* risk cost and MCTS time-vs-risk reward.
Guard reaction	Prioritize hide / Retreat+replan / Consider Echo	Changes candidate-action ordering and tactical reward, not world physics.
Emergency priority	Escape-first / Melee if adjacent / EMP if trapped	Changes utility overrides and scarce-tool thresholds.
Risk tolerance	Cautious / Balanced / Bold	Scales risk, noise, uncertainty and time terms; clamp to safe configuration bounds.

Starting presets: documented behavior, not secret buffs

PRESET	INITIAL BLOCK COMBINATION	TRADEOFF
GHOST	Safest / Hide / Escape-first / Cautious	Low detection but may be slow or trapped by conservative waiting.
BALANCED	Balanced / Retreat+replan / EMP if trapped / Balanced	Reacts flexibly; recommended baseline for tuning AI solvability.
RUSH	Shortest / Consider Echo / Melee if adjacent / Bold	Fast attempts with higher chance of contact, noise and capture.

Visible learn-by-testing loop



Sample report: "Captured at Research Junction; chose shortest exposed route after seeing Tracker; EMP still unused. MCTS preferred Advance (48 visits) over Retreat (11)." On retry, user changes Navigation to Safest and Guard Reaction to Retreat+replan. They can compare on the **same seed with NEXUS memory reset** to isolate policy changes.

Policy safety and persistence

- Save named JSON profiles with an ID, preset base, four block fields, policy version and optional user label. Never allow arbitrary executable code in config.
- Validate allowed enum values and numerical weight bounds. Apply policy only to companion decision-making, not guard HP, detection radius or map geometry.
- Report outcome, elapsed time, detections, gadgets, capture cause/sector, observed threat, MCTS visits/action and policy ID. Allow "Retry this seed" and "New challenge seed".

WHAT "TRAINING" MEANS HERE

The user performs manual iterative optimization. The robot does **not** persistently learn its own weights or train with ML/RL. MCTS planning happens during each run; it is not persistent model training.

14 / SCENE + UI SPECIFICATION

Menus, game HUD and results interface

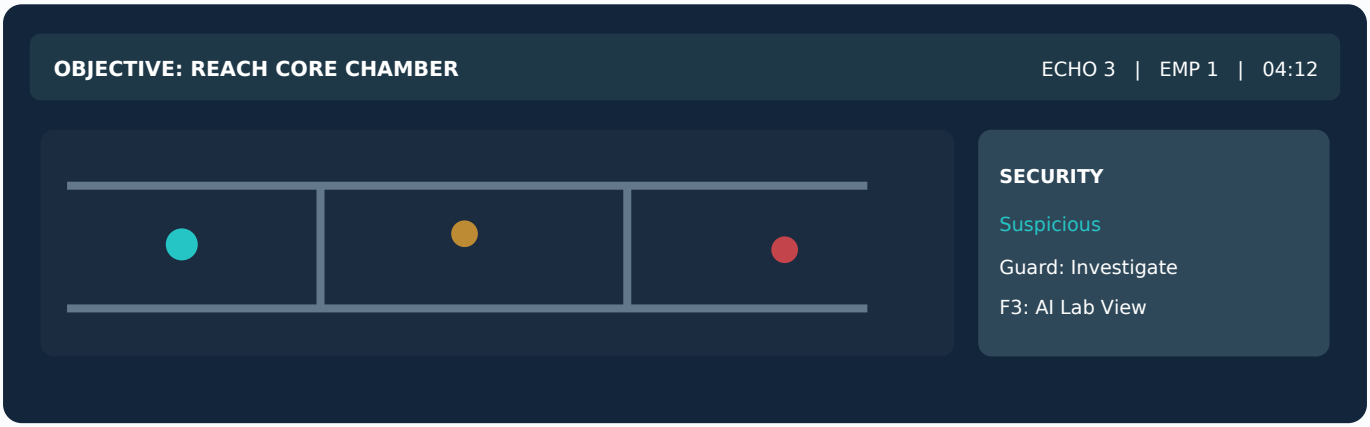
Clear AI vs AI access is mandatory for the lab project; make Companion Mode visible before the first playthrough.

Scene graph and navigation

01	02	03	04	05
MAIN MENU	MODE SELECT	WORKSHOP (AI only)	HEIST SCENE	RESULT / RETRY

Main Menu shows **Start Heist**, AI Lab / Algorithm Guide, Settings and Quit. Mode Select prominently exposes **Human vs AI** and **AI vs AI - Companion**. Human starts at Entry immediately. AI mode opens Workshop, then a deploy action starts the same map. Both routes end at identical result screens.

Gameplay HUD reference (not final pixels)



Workshop and Results wireframe content

BEFORE THE RUN

WORKSHOP PANEL
Preset tabs: Ghost / Balanced / Rush. Four drop-down blocks; policy summary, optional seed field, toggle Reset NEXUS Memory. Large "Deploy Companion" CTA.

AFTER THE RUN

RESULTS PANEL
WIN or CAPTURED; rank/achievement; time, detections, gadgets, location, decisive AI choice. Actions: Retry same seed / New seed / Edit Workshop / Menu.

Accessibility and responsive windows

Render at a logical reference size (e.g. 1280x720), scale UI for 1366x768 and 1920x1080, preserve aspect ratio, and expose windowed/fullscreen mode. Add pause, separate music/SFX sliders, reduced flashing, legible status icons as well as color, and subtitles for critical NEXUS messages.

15 / SYSTEMS DESIGN

Engineering architecture and state machine

Separate the deterministic simulation from rendering so AI experiments can run headlessly and real-time gameplay can share identical rules.

Runtime pipeline

01 **WorldState** → 02 **Filter observations** → 03 **Human or AI Actions** → 04 **Fixed simulation step** → 05 **Event log + render**

A top-level GameController manages MENU / MODE / WORKSHOP / PLAY / RESULT / EPILOGUE. The headless Simulation owns physics, entity inventories, guard vision, noise, mission progress, alarms, score and win/fail evaluation. Pygame handles input, presentation, animation and audio. Controllers return structured actions; no controller can mutate the world directly.

Recommended Python package boundaries

DIRECTORY	OWNERSHIP
game/core/	WorldState, entities, collision, fixed_tick, events, observation filters, action validation, mission FSM, scoring.
game/level/	Tiled JSON importer, object layers, navigation grid, LOS occlusion, sector/chokepoint graph, path caches.
game/ai/	BFS, A*, fuzzy, memory, guards, alpha-beta, MCTS, NEXUS director, companion controller and beliefs.
game/workshop/	Policy enums, presets, validation, profile save/load, run comparison.
game/ui/	Main menu, modes, workshop, HUD, result reports, Lab View, accessibility, camera and animation.
assets/ + data/	Textures, sprite atlases, SFX/music, Tiled maps, dialogue and default config.
tests/ + scripts/	Unit/integration tests, fixed-seed experiments, benchmark runner and platform packaging.

Deterministic tick order - all gameplay is resolved here

FIXED-TICK UPDATE CONTRACT

```
1. Read queued player input / AI choice from last Observation
2. Validate legality and resources; resolve movement + collisions
3. Execute interaction/ability actions; emit NoiseEvent/CORE_TAKEN
4. Update sensors (sight, acoustic reach) and memory
5. Update NEXUS orders, guard FSM and capture wind-ups
6. Evaluate global alert, mission win/loss, rank; append JSONL log
7. Render the committed state; deliver next filtered Observation
```

FAIRNESS / CODE REVIEW RULE

AI controllers receive immutable typed Observation objects; only Simulation has authoritative WorldState. MCTS rollout uses a copy of the agent belief state, **not** a hidden clone of the real world. Make tests fail if controllers access forbidden fields.

16 / IMPLEMENTABLE CONTRACTS

Data schemas, Tiled objects and configuration

Keep inputs explicit, serializable and testable; never hide gameplay constants in rendering code.

Core data interfaces (indicative JSON / Python fields)

DATA CONTRACTS

```
WorldState: tick, phase, actors[], grid, doors, core_held,
            noise_events[], guard_memories[], nexus_memory, seed
Observation: agent_id, tick, visible_entities[], heard_events[],
             shared_reports[], map_static, phase, own_status, inventory
Action: actor_id, kind[MOVE|WAIT|INTERACT|ECHO|EMP|MELEE|HIDE],
        target_tile?, direction?, tick_issued
SecurityReport: source_guard, report_kind, tile, tick, confidence
RunResult: mode, seed, outcome, seconds, detections, gadgets,
           policy_id?, capture_by?, sector?, grade, decisive_events[]
```

Suggested map object types and required properties

TILED LAYER / OBJECT	FIELDS / SEMANTICS
Ground + Decor	Render-only tile layers, optional foreground above actors.
Solid + Opaque	Tile flags: movement_blocked, vision_blocked, sound_attenuation.
ZonePolygon	id, name, ambience, sector_graph_node.
Spawn / Exit / Core	Unique stable object IDs; core once-only interaction; win checks own core inventory.
HideSpot / Door / Prop	type, collider, use_range, initial_state, sound_event_kind, animation_id.
PatrolPath / Chokepoint	guard assignment and ordered nodes; interceptor abstract route node and links.

Sample AI Workshop policy file

VALIDATED CONFIG / EXAMPLE

```
{
  "id": "custom_01", "base": "BALANCED", "policy_version": 1,
  "navigation": "SAFEST", "guard_reaction": "RETREAT",
  "emergency": "EMP_IF_TRAPPED", "risk": "CAUTIOUS",
  "mcts_simulations": 64, "seed": 1042,
  "reset_nexus_memory": true
}
```

Persistence: settings.json, policies.json, run_history.json and JSONL telemetry in a user-writable application data directory. Do not write saves into read-only packaged application folders. Include deterministic schema versions and backward-safe defaults.

17 / LAB EXPERIMENTS + ACCEPTANCE

Scoring, algorithm evidence and quality gates

Success must be measured as an autonomous adversarial game, not a scripted demonstration.

Outcome grading / proposed first tuning pass

COMPONENT	INITIAL SCORE RULE
Core extraction	60 points only when core is held and extraction reached. Failure is always F.
Stealth	Up to 20: penalize confirmed detections and repeatedly unnecessary noisy confrontations, not normal Echo use.
Elapsed time	Up to 10: baseline curve from completed-run playtests; never a hard mission deadline.
Resource discipline	Up to 10: reward avoiding needless EMP/melee; do not punish needed tactical Echo use by default.
Ranks / achievements	S >=90, A >=75, B >=60, C otherwise on win. Ghost = no confirmed sighting; Speedrunner = tested threshold; AI Master = Companion victory.

Concrete acceptance scenarios

TEST ID	GIVEN / WHEN	REQUIRED RESULT
T01	Human mode and AI mode on fresh installation	Both launch without prior success or unlock.
T02	Companion reaches core with no detected enemies	CORE_TAKEN once; lockdown; extraction objective.
T03	Guard hears glass behind opaque wall	Investigate sound tile; no hidden-player coordinates or confirmed contact.
T04	Intruder breaks line of sight after chase	Tracker searches last seen tile; memory eventually expires.
T05	EMP during capture wind-up before resolution	Guard disabled, grab canceled; resource consumed; noise emitted.
T06	Interceptor receives plausible route reports	Minimax selects legal post; physically travels; alpha-beta nodes logged.
T07	Seed fixed, NEXUS memory reset, policy changed	Deterministic same-config replay; differing policies change decisions in test scenarios.
T08	Core stolen, escape attempted without/with core	Lockdown occurs; only carrier extraction succeeds.

Academic evidence and experiments

Run a reproducible fixed-seed suite (e.g. 20-30 scenarios covering patrol timings, chokepoints and observed threats) for Ghost/Balanced/Rush and one user-tuned policy. Compare win rate, time, detections, gadget spend, A* nodes/path cost, Sentinel fuzzy urgency values, MCTS action visits/rewards and alpha-beta evaluated/pruned nodes. Also compare minimax with pruning switched off on the same small tactical states.

NON-NEGOTIABLE PASS CONDITIONS

Two genuinely opposing AI sides; companion autonomous win and fail possible; legitimate observation limits; algorithms actually executed and logged; meaningful Workshop variation; no impossible map; fair pursuit/capture; working result screens; source and video ready for lab showcase.

18 / PRODUCTION PLAN AND DEFINITION OF DONE

Four-week execution, assets and release

A practical handoff roadmap for two developers. Prioritize algorithm correctness and one playable mission over ornamental features.

Delivery milestones and ownership

WEEK	DEVELOPER A / ENGINE + WORLD	DEVELOPER B / AI + INTERFACE
01 Playable heist	Tiled map, collision, player, core/exit, noise, hiding, capture, lockdown; first packaged smoke test.	Sentinel BFS/FSM, perception/observation API, HUD/menu prototype, shared data contracts.
02 AI vs AI active	Headless deterministic runtime, logging, seeded replay, Tracker A*, movement/action validators.	Companion FSM + heuristic A* + utility, Interceptor minimax, NEXUS rules and two-mode UI.
03 AI proof + tools	Echo path buffer, EMP/melee, props, SFX/art, capture/alert refinements and path bugs.	Fuzzy membership rules, bounded MCTS, Workshop presets, result report, Lab View metrics.
04 Ship	Balance gameplay, QA, capture demos, platform ZIPs and itch.io page.	Verify algorithm telemetry, fixed-seed comparisons, README, GitHub tests, lab presentation/video.

Asset, platform and release checklist

- Assets: 5-room tileset, 4 actor silhouettes, alert states, 2 prop sprites, energy-core VFX, Echo/EMP effects, 6-10 gameplay SFX and one loopable ambience/music layer. Track CC0/permissive license and attribution in ASSETS.md.
- Technology: Python + pygame-ce, Tiled JSON, pytest, deterministic seed runner, PyInstaller one-folder platform builds, GitHub Actions where practical.
- Desktop target: Windows x64 and macOS tested hardware; Linux x64 only if clean-machine-tested. Mac unsigned Gatekeeper instructions in README.
- GitHub: source, install/run guide, license/credits, screenshots, demo video, algorithm descriptions and test command. itch.io: free ZIP files, concise pitch, controls and troubleshooting.

Scope control and final acceptance

OPTIONAL POLISH

CUT FIRST IF LATE
Extra props, lighting upgrades, elaborate cinematics, extra sound layers, advanced Lab View visualizations, untested Linux package.

CORE IDENTITY

DO NOT CUT
Complete start-to-extract heist, first-launch AI vs AI, real MCTS/fuzzy/A*/minimax usage, fair observation, Workshop impact, replayable seeded tests, win/fail flow.

BUILD COMPLETE WHEN

A new user can install the game, choose either mode, complete or fail the same Chapter One mission, understand why, modify the autonomous companion, replay and compare strategies; the team can show real logged AI decisions and submit a GitHub link plus gameplay video.